Pre-Analysis Plan

Decisions and assumptions common to all analyses

For causal interpretation of our results, the identifying assumption is that conditional on having children, the sex of the child is random.

Independent variable

For all analyses, we construct an indicator “has daughter” that is equal to 1 if an individual has at least one daughter and 0 if an individual has at least one son and no daughter. The indicator is missing for individuals who have no children.

Exclusion from the analysis sample

We restrict our analysis to individuals with at least one child and who reside in private households. Observations in which the reported number of births does not align with the number of children indicated in the survey are excluded, as such discrepancies may reflect measurement error. All available time periods in which the dependent variable is observed are included in the analysis.

We clean all variables that should take positive values only for valid responses by setting negative values (e.g., labor income = -2, years of schooling = -2) to missing. We do not impute missing values for either dependent or independent variables.

Additionally, we exclude observations involving same-sex couples and nonbinary individuals from the analysis.

H1

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*

Dependent variable: We measure attitudes with six SOEP items that we combine into a Gender Equality Index (GEI; varname = gender_eq_index):

Question / Year	2012	2018
Children below the age of 6 suffer if their mother works [lowest = Completely Disagree; highest = Fully agree]	plh0298_v1	plh0298_v2
Children below the age of 3 suffer if their mother works [lowest value = Completely Disagree; highest value = Fully agree]	plh0302_v1	plh0302_v2
It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally [lowest = Completely Disagree; highest = Fully agree]	plh0308_v1	plh0308_v2

Because for plh0298 and plh0302, higher scores indicate less egalitarian attitudes, we reverse these items so that higher values mean more egalitarian views. The items are coded on different scales in different waves (v1: 1-4; v2: 1-7), such that we standardize all six items. For each wave, we combine the standardized items into one equally-weighted index. The approach follows Hamjediers (2021), who used the same items to build a cross-sectional index in the 2018 wave.

Control variables: Attitudes for gender equality have been found to be associated with respondent gender (McDaniel, 2008), Socio-Economic status proxied by education (Davis, 2007) and income (Beyerlein, 2007), age (Davis, 2007) and marriage status (Kaufman et al., 2017), so we also include the following control variables

- Control 1.1: Gender of Respondent [sex]
    - We recode sex to be a male dummy (equal to 1 for men and equal to 0 for women).
- Control 1.2: Education in years [pgbilzeit]
    - Measured as years of schooling from 7 (no degree) to 15 (university degree)
- Control 1.3: Individual Gross Labor Income [pglabgro]
    - Represents the imputed gross labor income in the previous month generated for all SOEP respondents who are employed in a main job in each respective wave. Interviewees should consider overtime payments but no irregular one-time payments such as holidays or bonuses.

    - Respondents classified as unemployed, based on their occupational status in variable [pgstib], are manually assigned a gross labor income of 0.
- Control 1.4: Respondent Marital status [pgfamstd]
    - We use indicators for the different situations with marriage as the base category.
- Control 1.5: Age
    - Calculated by subtracting respondents' birth year [gebjahr] from the survey year [syear]

Further, we also control for

- Control 2.1: Total Number of Children at time of Survey [num_kids]
    - Num_kids represents the number of children identified in the survey, determined by counting the birth years of children that precede the survey year.
- Control 2.2: Migration Background indicators [migback]
    - The variable contains information on respondents' migration background for all persons who have ever been a part of a SOEP household (direct/indirect/no background). We use no migration background as the reference category.
- Control 2.3: Household Size [hhsize]
    - Measured as the number of individuals living in the same household as the survey respondents.
- Control 2.4: Regional indicator [sampreg]
    - The variable SAMPREG indicates whether a HH was located in West (=1) or East Germany (=2) in the respective year. This variable approximates also former regional differences within Berlin according to zip-codes. We recode the variable as an East-Germany dummy that is equal to 1 for households in East Germany and 0 otherwise.
- Control 2.5: 2018 indicator [syear]
    - Due to dependent variable only being observed in 2012 and 2018, we have a 2018 dummy that is equal to 1 for 2018 and 0 for 2012.
- Control 2.6: Interaction between year and regional dummies

Further inclusion/exclusion criteria: none

Analysis:

We use OLS to estimate the following equation:

$$\mathrm{GEI}_{i,t} = \alpha + \beta \mathrm{D}_{i,t} + \gamma' \mathrm{X}_{i,t} + \varepsilon_{i,t}$$

where $GEI_{i,t}$ is the gender equality index of individual *i* in year *t*, $D_{i,t}$ is the indicator for having at least one daughter, $X_{i,t}$ is a vector of the above-described control variables. The main coefficient of interest is β, where $\hat{\beta} > 0$ is evidence in support of H1. We cluster the robustly estimated standard errors at the household level since individuals cohabiting might influence each other's attitudes.

I**nterpretation of results**:

We are interested in $\hat{\beta}$, the estimated effect of having at least one daughter on the GEI. Under the null hypothesis (no effect of daughters on gender egalitarian attitudes), this coefficient is zero. We conduct a two-sided t-test at the 5% significance level. If the estimated coefficient $\hat{\beta}$ is significantly larger than zero, we reject the null hypothesis and conclude that having at least one daughter increases gender egalitarian attitudes (as compared to having only sons), consistent with H1.

H2

H2. *Having daughters leads to more left-leaning political preferences.*

Dependent variable: The outcome variable is "leftleaning", that we coded on an 8-point scale from 1 (most right-wing) to 8 (most left-wing) as follows, based on plh0012_h ("Which party do you lean toward?"):

Score	Party
1	NPD/Republikaner
2	AfD
3	CDU/CSU
4	FDP
5	SPD
6	Bündnis 90/Die Grünen
7	Piratenpartei
8	Die Linke

The sorting of parties from left to right is based on Tangian (2022), who used principal-component and factor analyses to define with this mapping.

Control variables: As political preferences have been found to be associated with gender (Edlund & Pande, 2002), age (Oswald & Powdthavee, 2010), education (Edlund & Pande, 2002), income (Edlund & Pande, 2002), marriage status (Edlund & Pande, 2002), number of children (Borrell-Porta et al., 2019), migration background (Jessen, 2022) and regional preferences as well as varying trends over time, we also include the following control variables

- Control 1.1: Gender of Respondent [sex]
    - We recode sex to be a male dummy (equal to 1 for men and equal to 0 for women).
- Control 1.2: Education in years [pgbilzeit]
    - Measured as years of schooling from 7 (no degree) to 15 (university degree)
- Control 1.3: Individual Gross Labor Income [pglabgro]
    - Represents the imputed gross labor income in the previous month generated for all SOEP respondents who are employed in a main job in each respective wave. Interviewees should consider overtime payments but no irregular one-time payments such as holiday or bonuses.
    - Respondents classified as unemployed, based on their occupational status in variable [pgstib], are manually assigned a gross labor income of 0.
- Control 1.4: Respondent Marital status [pgfamstd]

    - We use indicators for the different situations with marriage as the base category.
- Control 1.5: Age
    - Calculated by subtracting respondents' birth year [gebjahr] from the survey year [syear]

Further, we also control for

- Control 2.1: Total Number of Children at time of Survey [num_kids]
    - Num_kids represents the number of children identified in the survey, determined by counting the birth years of children that precede the survey year.
- Control 2.2: Migration background indicators [migback]
    - The variable contains information on respondents' migration background for all persons who have ever been a part of a SOEP household (direct/indirect/no background). We use no migration background as the reference category.
- Control 2.3: Household Size [hhsize]
    - Measured as the number of individuals living in the same household as the survey respondents.
- Control 2.4: Regional indicator [sampreg]
    - The variable SAMPREG indicates whether a HH was located in West (=1) or East Germany (=2) in the respective year. This variable approximates also former regional differences within Berlin according to zip-codes. We recode the variable as an East-Germany dummy that is equal to 1 for households in East Germany and 0 otherwise.
- Control 2.5: Time dummies [syear]
    - We include time indicators.
- Control 2.6: Interaction between time and regional indicators.

Inclusion/exclusion criteria: We exclude individuals who indicate they lean toward parties other than the ones listed above or a mix of those, as these responses are hard to classify and may introduce measurement error.

Analysis:

Using ordered logistic regression, we estimate the equation:

$$leftleaning_{\mathrm{i,t}} = \alpha + \beta \mathrm{D}_{\mathrm{i,t}} + \gamma' \mathrm{X}_{\mathrm{i,t}} + \varepsilon_{\mathrm{i,t}},$$

where $leftleaning_{\mathrm{i,t}}$ is the observed left-right placement of respondent *i* at time *t*, $D_{i,t}$ is the indicator for having at least one daughter, $X_{i,t}$ is a vector of the above-described control variables. The main coefficient of interest is β, where $\hat{\beta} > 0$ is evidence in

support of H2. We cluster the robustly estimated standard errors at the household level since individuals cohabiting might influence each other's political preferences.

Interpretation of results:

We are interested in the estimated effect of having at least one daughter, $\hat{\beta}$, on *leftleaning*. Under the null hypothesis (no effect of daughters on political preferences), this coefficient is zero. We conduct a two-sided z-test at the 5% significance level. If the estimated coefficient $\hat{\beta}$ is significantly larger than zero, we reject the null hypothesis and conclude that having at least one daughter (as compared to none) increases left leaning political preferences, consistent with H2.

H3

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*

Dependent variable: We use a behavioral measure of gender equality in the household. For this analysis, we consider the household head and their (opposite-sex) partner. The outcome variable is female unpaid labor share [varname = fem_unpaid_labor_share], defined as the share of weekly unpaid work hours in the household performed by the woman. The idea and procedure is based on Jessen (2022). The variable fem_unpaid_labor_share is calculated by using the following SOEP items:

	Weekday	Saturday	Sunday
Housework	pli0012_h	pli0016_h	pli0043_h
Childcare	pli0019_h	pli0022_h	pli0044_h
Repairs	pli0031_h	pli0049_h	Not available

In these items, individuals indicate how many hours per week they use for the respective category of unpaid labor ("How many hours do you spend on the following activities on a typical weekday, Saturday, and Sunday?"). After multiplying the number of unpaid labor hours by 5 for the weekday observations, the number of hours of unpaid labor is them summed by individual and by household, and the female share of unpaid labor is calculated as the share of unpaid labor hours within the sum of unpaid labor hours by couple which is identified by personal- and partner ids provided in the dataset:

$$fem_unpaid_labor_share_h = \frac{unpaid_labor_{female}}{unpaid_labor_{female} + unpaid_labor_{male}}$$

Control variables: As gender equality measured by share of unpaid household labor has been found to be associated with Socio-Economic status proxied by education (Jessen, 2022) and household income (Jessen, 2022), age (Jessen, 2022), marriage status (Jessen, 2022), we also include the following control variables

- Control 1.1.1: Difference in Education in years between partners [edu_diff_mf]
    - Calculated by subtracting the number of years of education of the female partner from the years of education of the male partner [pgbilzeit]
- Control 1.1.2: Average Education of Partners [avg_edu]
    - Calculated by averaging the female and male partners' individual years of education [pgbilzeit]
- Control 1.2.1: Difference in Age in years between partners [age_diff_mf]

    - Calculated by subtracting the age of the female partner from the age of the male partner [based on age, calculated by subtracting birth year from survey year]
- Control 1.2.2: Average Age of Partners [avg_age]
    - Calculated by averaging the age of the female partner and the male partner [based on age, calculated by subtracting birth year from survey year]
- Control 1.3: Household Gross Labor Income [hghinc]
    - Contains the current monthly net household income asked for in the household questionnaire (in EUR).
- Control 1.4: Marital status [pgfamstd]
    - We use indicators for the different situations with marriage as the base category.

Further, we also control for

- Control 2.1: Total Number of Children at time of Survey [num_kids]
    - Num_kids represents the number of children identified in the survey, determined by counting the birth years of children that precede the survey year.
- Control 2.2: Migration Background indicators [migback]
    - The variable contains information on respondents' migration background for all persons who have ever been a part of a SOEP household (direct/indirect/no background). We use no migration background as the reference category.
      If Partners have different Migration Backgrounds, we assign
        - indirect migration backgrounds if at least one partner has an indirect migration background and no partner has a direct migration background
        - direct migration backgrounds if at least one partner has a direct migration background
- Control 2.3: Regional indicator [sampreg]
    - The variable SAMPREG indicates whether a HH was located in West (=1) or East Germany (=2) in the respective year. This variable approximates also former regional differences within Berlin according to zip-codes. We recode the variable as an East-Germany dummy that is equal to 1 for households in East Germany and 0 otherwise.
- Control 2.4: Year indicators [syear]
- Control 2.5: Interaction of year and regional indicators

Inclusion/exclusion criteria: We focus on household heads and their partners, i.e., we exclude individuals without partners or individuals whose partners cannot be identified in the SOEP dataset by the partner id [parid].

Analysis:

We use OLS to estimate the equation:

$$fem_unpaid_labor_share_{h,t} = \alpha_h + \beta D_{h,t} + \gamma' X_{h,t} + \varepsilon_{h,t}$$

where $fem_unpaid_labor_share_{h,t}$ is the female share of unpaid labor of household *h*, D_h is the indicator for having at least one daughter, and X_i is a vector of the above-described control variables. The main coefficient of interest is β_D, where $\widehat{\beta_D} < 0$ is evidence in support of H3. We robustly estimate standard errors.

Interpretation of results:

We are interested in $\hat{\beta}$, the estimated effect of having at least one daughter on *female unpaid labor share*. Under the null hypothesis (no effect of daughters on gender equality in the household measured by behavior), this coefficient is zero. We conduct a two-sided t-test at the 5% significance level. If the estimated coefficient $\hat{\beta}$ is significantly smaller than zero, we reject the null hypothesis and conclude that having daughters increases gender equality in the household measured by behavior, consistent with H3.

H4

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*

Dependent variable: We measure attitudes with six SOEP items that we combine into a Gender Equality Index (GEI; varname = gender_eq_index):

Question / Year	2012	2018
Children below the age of 6 suffer if their mother works [lowest = Completely Disagree; highest = Fully agree]	plh0298_v1	plh0298_v2
Children below the age of 3 suffer if their mother works [lowest value = Completely Disagree; highest value = Fully agree]	plh0302_v1	plh0302_v2
It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally [lowest = Completely Disagree; highest = Fully agree]	plh0308_v1	plh0308_v2

Because for plh0298 and plh0302, higher scores indicate less egalitarian attitudes, we reverse these items so that higher values mean more egalitarian views. The items are coded on different scales in different waves (v1: 1-4; v2: 1-7), such that we standardize all six items. For each wave, we combine the standardized items into one equally-weighted index. The approach follows Hamjediers (2021), who used the same items to build a cross-sectional index in the 2018 wave.

As attitudes for gender equality have been found to be associated with respondent gender (McDaniel, 2008), Socio-Economic status proxied by education (Davis, 2007) and income (Beyerlein, 2007), age (Davis, 2007) and marriage status (Kaufman et al., 2017), we also include the following control variables

- Control 1.1: Gender of Respondent [sex]
    - We recode sex to be a male dummy (equal to 1 for men and equal to 0 for women).
- Control 1.2: Education in years [pgbilzeit]
    - Measured as years of schooling from 7 (no degree) to 15 (university degree)
- Control 1.3: Individual Gross Labor Income [pglabgro]
    - Represents the imputed gross labor income in the previous month generated for all SOEP respondents who are employed in a main job in each respective wave. Interviewees should consider overtime payments but no irregular one-time payments such as holiday or bonuses.

    - Respondents classified as unemployed, based on their occupational status in variable [pgstib], are manually assigned a gross labor income of 0.
- Control 1.4: Respondent Marital status [pgfamstd]
    - We use indicators for the different situations with marriage as the base category.
- Control 1.5: Age
    - Calculated by subtracting the respondent's birth year [gebjahr] from the survey year [syear]

Further, we also control for

- Control 2.1: Total Number of Children at time of Survey [num_kids]
    - Num_kids represents the number of children identified in the survey, determined by counting the birth years of children that precede the survey year.
- Control 2.2: Migration Background indicators [migback]
    - The variable contains information on respondents' migration background for all persons who have ever been a part of a SOEP household (direct/indirect/no background). We use no migration background as the reference category.
- Control 2.3: Household Size [hhsize]
    - Measured as number of individuals living in the same household as the survey respondents.
- Control 2.4: Regional indicator [sampreg]
    - The variable SAMPREG indicates whether a HH was located in West (=1) or East Germany (=2) in the respective year. This variable approximates also former regional differences within Berlin according to zip-codes. We recode the variable as an East-Germany dummy that is equal to 1 for households in East Germany and 0 otherwise.
- Control 2.5: Time indicator [syear]
    - Due to dependent variable only being observed in 2012 and 2018, we have a 2018 dummy that is equal to 1 for 2018 and 0 for 2012.
- Control 2.6: Interaction between year and regional dummies.

Additional inclusion/exclusion criteria: none

Analysis:

We use OLS to estimate the following equation:

$$\mathrm{GEI_{i,t}} = \alpha + \beta_D \mathrm{D_{i,t}} + \beta_M \mathrm{M_i} + \beta_{DM}\left(\mathrm{D_{i,t}} \times \mathrm{M_i}\right) + \gamma' \mathrm{X_{i,t}} + \varepsilon_{\mathrm{i,t}}$$

where $GEI_{i,h}$ is the gender equality index of individual *i* at time *t*, $D_{i,t}$ is the indicator for having at least one daughter, M_i is a dummy variable taking value 1 for male respondents and $X_{i,t}$ is a vector of the above-described control variables. The main coefficient of interest is the coefficient on the interaction term, β_{DM}, where $\hat{\beta}_{DM} > 0$ is evidence in support of H4. We cluster the robustly estimated standard errors at the household level since individuals cohabiting might influence each other's attitudes.

Interpretation of results:

We are interested in $\hat{\beta}_{DM}$, the estimated differential effect of having at least one daughter on the GEI for men. Under the null hypothesis (no differential effect of daughters on gender egalitarian attitudes for men), this coefficient is indistinguishable from zero. We conduct a two-sided t-test at the 5% significance level. If the estimated coefficient $\hat{\beta}_{DM}$ is significantly larger than zero, we reject the null hypothesis and conclude that having at least one daughter has a differential effect on GEI for men, consistent with H4.

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